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B.Tech. Degree VII Semester Examination November 2018

CS 15-1703 ANALYSIS AND DESIGN OF ALGORITHMS (2015 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A (Answer ALL questions)

(10 × 2 = 20)

- I. (a) Let $w(n)$ and $a(n)$ denote respectively, the worst case and average case running time of an algorithm executed on input size n . Given that $a(n) = \Omega(w(n))$. Do you agree with this? Justify your answer.
- (b) Solve the recurrence relation $T(n) = \sqrt{2} T\left(\frac{n}{2}\right) + \sqrt{n}$, $T(1) = 1$, when $n = 2^{2k}$ for some $k \geq 0$, by using substitution method.
- (c) Using masters theorem solve $T(n) = 4T\left(\frac{n}{2}\right) + n^2$.
- (d) Differentiate between Binomial heaps and Fibonacci heaps using suitable example.
- (e) Estimate worst case, best case and average case complexity of linear search.
- (f) Write a note on amortized analysis.
- (g) Demonstrate DFS algorithm using suitable example.
- (h) Give Warshall algorithm for transitive closure of a directed graph.
- (i) Write notes on the P, NP, NP hard and NP complete complexity classes and also discuss on whether $P \equiv NP$.
- (j) Write a note on algorithm for parallel computers.

PART B

(4 × 10 = 40)

- II. (a) Solve the recurrence $T(n) = 2T\left(\frac{n}{3}\right) + n^2 \log n$ using Masters theorem. (5)
 - (b) Differentiate between greedy and dynamic programming. (5)
- OR
- III. (a) Solve the recurrence $T(n) = 3T\left(\frac{n}{4}\right) + \Theta(n^2)$ substitution method. (6)
 - (b) Using Masters theorem solve $T(n) = 3T\left(\frac{n}{3}\right) + \sqrt{n} \log^2 n$. (4)

(P.T.O.)

- IV. Write down an algorithm for quick sort and analyze its best case and worst case complexity. Give a small example also. (10)

OR

- V. (a) Define binomial heap. Write an algorithm for inserting a node in a binomial heap and also give its complexity. (6)
(b) Compare the performance of linear search and binary search. (4)

- VI. (a) Write and demonstrate algorithm for finding all pair shortest path in a weighted graph and also give its complexity. (6)
(b) Demonstrate matrix chain multiplication using suitable example. (4)

OR

- VII. (a) What is a strongly connected component of a diagraph? Explain algorithm for finding strongly connected components with an example. (6)
(b) Give an algorithm for constructing optimal binary search tree. (4)

- VIII. Give Knuth-Morris Pratt algorithm for string matching. Also give its complexity. (10)

OR

- IX. (a) Explain travelling sales man problem using branch and bound with suitable example. (6)
(b) Write a note on graph coloring problem. (4)
